

INDEX

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MODULE-I



Page No :

Date : .../.../...

(*) Data Mining : process of extracting useful patterns (knowledge) from huge sets of data. It is a multidisciplinary field that uses ML, statistics and AI to extract info. to evaluate future events' probability.

* Pros	* Cons
• enables orgs. to obtain knowledge based data Customer seg, Customer purchase pattern, Target marketing • helps in decision making process • helps in making lucrative changes in operation and production	• selecting right data mining tools is challenging • data mining analytics s/w are difficult to operate and need advanced learning & training to work on. • Privacy issues

Customer profiling, risk management, finance planning, asset evaluation, resource planning

* Applications

- i) Fraud detection in Messages & Emails
- ii) ~~Law~~ Detection in Law Enforcement
- iii) Weather Forecasting in Meteorological department
- iv) Market Based Analysis by Business Companies
- v) Healthcare by Medical organizations.

* Challenges in Data Mining

- i) Real world data is unstructured, heterogeneous, incomplete and noisy
- ii) Managing heterogeneous data is a tedious task
- iii) Serious issues of privacy and security.
- iv) Representing data visually is also difficult.
- v) It is also difficult to create a centralized repository to store distributed data
- vi) There are many algorithms to mine the data. So, if the designed algorithm & techniques are not upto the mark, the efficiency of data mining is reduced severely.

* KDD

Knowledge Discovery in Databases :- process of discovering knowledge in data that was previously unknown and is potentially valuable. It is an iterative process and requires multiple iterations of some steps for data mining.

Data Cleaning - removal of noisy data & irrelevant data

- (a) missing values (b) noise - random or variance error
- (c) data discrepancy detection and transformation tools
- (d) smooth outliers (e) resolve inconsistencies

Data Integration - heterogeneous data sources are merged into single source of data

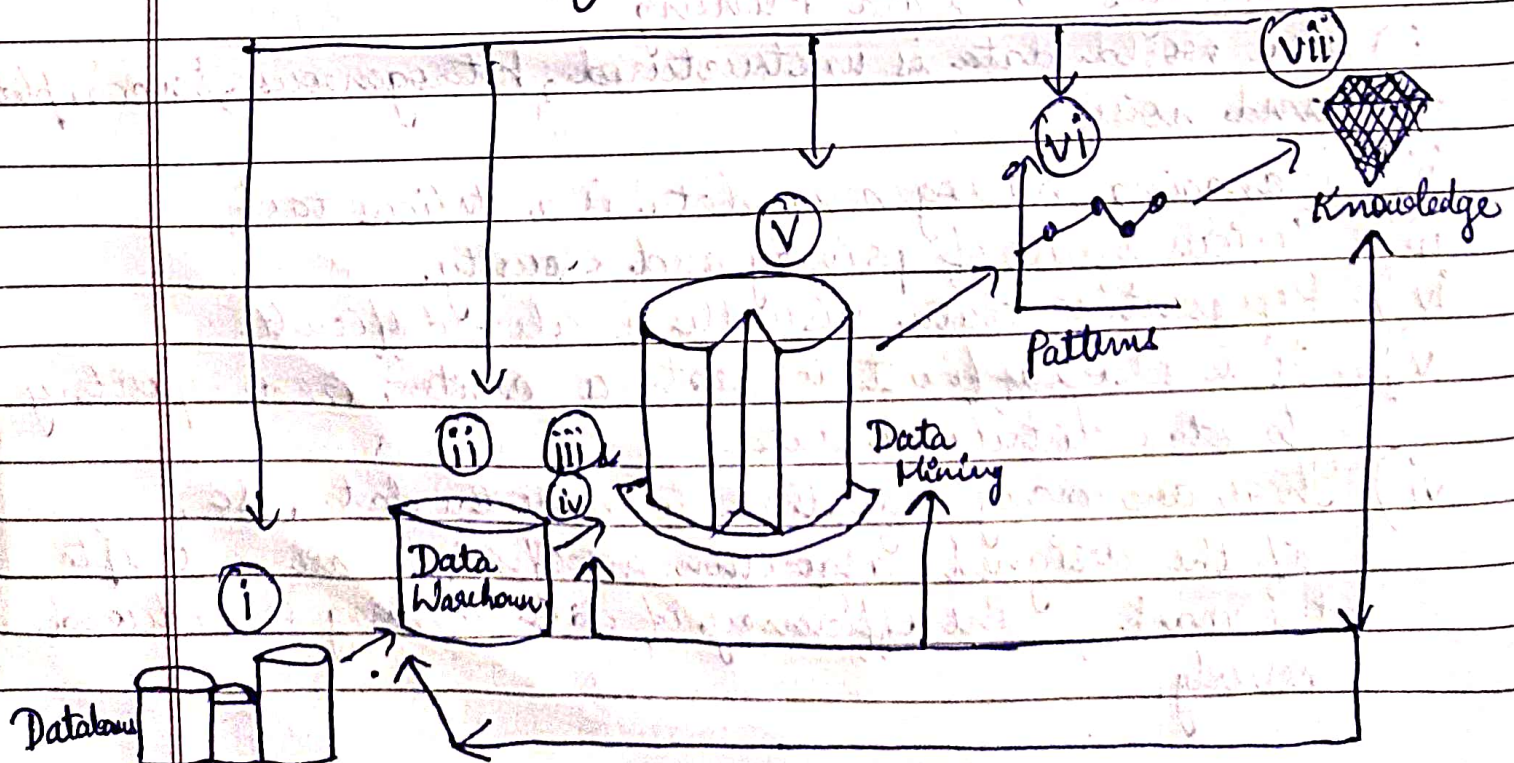
Data Selection - relevant data is retrieved from database.

Data Transformation - retrieved data is transformed into suitable form for data mining

Data mining - techniques are applied to this transformed data to extract patterns

Pattern Evaluation - identifying patterns and summarizing and visualizing and scoring them

Knowledge Representation - presenting results in a way that is meaningful and can be used for decision making

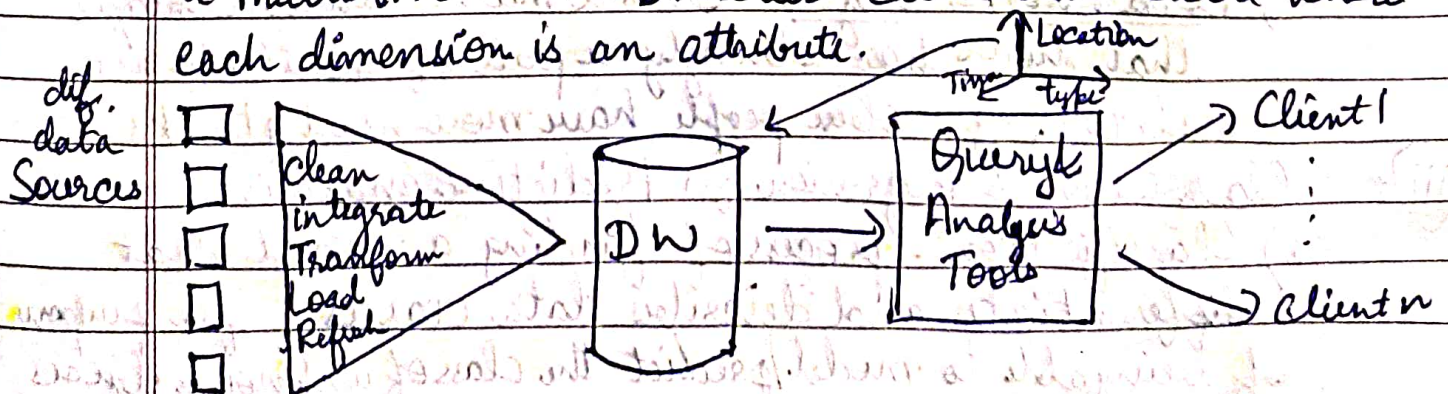


	Pros	Cons
* i) ii) iii) iv) v)	Improves decision making increases efficiency Better customer service fraud detection predictive modeling	i) Privacy concerns ii) high cost iii) Unintended results like bias iv) High Quality data dependant v) Overfitting

* Difference b/w KDD & Data Mining	
<u>KDD</u>	<u>Data Mining</u>
Discovers Knowledge from data	Patterns from data
A field in itself	An important step of KDD
Uses steps mentioned before	algorithms such as decision trees, neural network ML, association rules and classification clustering

(*) What kinds of data can be mined?

- i) Database Data (RDBMS) :- It is a set of tables. Tables have rows and columns. While mining databases, we search for trends or data patterns, like analysing customer data to predict credit risks of new customers, sales data.
- ii) Data Warehouse Data :- Datawarehouse is a warehouse / repository of information collected from multiple sources, stored under a unified schema. This schema is defined as a multidimensional structure called data cube where each dimension is an attribute.



iii) Data for Transactional Database :- A transactional database is a DBMS that can read/write rows very quickly while protecting the integrity they contain.

iv) Other types of data :- data streams, sequence data, spatial data, webdata, multimedia, hypertext etc.

* What kind of Patterns can be mined?

→ It is determined by data mining functionalities.

i) Class / Concept Description :- data entries can be associated with classes or concepts which need to be described.

It can be done through

a) Classification ^{Characterization} :- summarizing general characteristics or features of a target class of data. Eg:- generating a profile of CS students with high GPA.

b) Discrimination :- comparing general characteristics of target data with general characteristics of objects from one or a set of contrasting classes. Eg:- comparing profile of students with High GPA's to that of low GPA's to get 75% CS students having high GPAs & 65% students are from other branches.

ii) Mining frequent patterns, associations & correlations :-

a) frequent patterns - things which are found most commonly in data like substructure, subsequence

b) association analysis - way of identifying relations b/w various items. eg:- items purchased together freq.

c) correlation analysis :- It is a mathematical technique that shows how strongly a pair of attributes are related, eg:- obese people have more heart attacks.

iii) Classification & Regression for Predictive analysis

a) Classification - process of finding a model that differentiates and describes data classes, for purpose of being able to model/predict the class of unknown objects

b) Regression :- process to model continuous-valued functions, to predict missing or unavailable numerical data using previous set of data.

iv) Cluster Analysis :- analysing data without considering class labels, thus objects are grouped based on the principle of maximising intraclass similarity and minimising ^{b/w 2 diff} interclass similarity. Each cluster is viewed as a class of objects, can facilitate taxonomy formation, i.e. organising observations into a hierarchy of classes & that group classes together.

v) Outlier Analysis (Anomaly Mining) :- analysis of ~~object~~ data that do not comply with general model of data. Eg:- fraud detection based on a large data set of credit card transactions.

*

Which technologies are used?

- i) Statistics :- studies the collection, analysis, interpretation or explanation and presentation of data with the help of statistical model that describes the behaviour of objects in the target class in terms of random variables and their associated probability distributions.
- ii) Machine Learning :- It investigates how computers can learn or improve their performance based on data. They should automatically learn to recognize complex patterns and make intelligent decisions based on data.
- a) Supervised learning with the help of training set
→ Classification : Postal code Recognition.
- b) Unsupervised learning without training set → Clustering
eg :- image recognition
- c) Semi-supervised learning :- using both labeled and unlabeled examples when learning a model.
- d) Active learning :- Human interactive learning
- iii) Database Systems & Data Warehouse :-
- iv) Information Retrieval :- searching documents for information assuming it is unstructured and queries are formed mainly by keywords that do not have complex structure.

*

Which applications are targeted?

- • Web Search • E-commerce
- Business Intelligence • Supermarkets
- Banking & Finance • Crime Investigation
- Manufacturing • Bioinformatics
- Customer Support
- Communications
- Education

Issues in Data Mining

- * i) The rate at which data (from which knowledge can be extracted) is being generated far exceeds the ability to extract useful knowledge from them to facilitate decision making and to better understand the changing environment.
- ii) It is a great challenge how to utilize existing data mining techniques and to create new techniques to effectively exploit rich patterns embedded in the 'new' datasets. Because they can add complexities to such task.
- iii) Scalability :- As size of dataset increases, time and computational resources required to perform mining increases. Moreover, algorithms must be able to handle streaming data. To handle this, use distributed computing to distribute data into multiple nodes.
- iv) Interpretability :- algorithms produce complex models that are difficult to interpret. Solution - use visualization.
- v) Data Quality :- The accuracy, completeness and consistency affects mining. The data may have error, omissions, duplicates, noise, inconsistencies or outliers that can affect the result. Solution: data cleaning and preprocessing.
- vi) Data Complexity :- due to vast amount of data generated by dif. srcs. and in dif. formats, it is difficult to process, analyse, understand and store them. Solution: clustering, classification, association rule mining.
- vii) Data Privacy and Security :- As more data is stored, risk of data breaches and cyber attacks increases. Solution :- encryption and anonymization.

Data Object

'Represents an entity which is described by attributes.
eg:- Table.

Attribute

data-field representing a characteristic or feature of data object.

→ **Nominal Attribute** - symbols or names of things which have values not having meaningful order.
eg:- Occupation → teacher, dentist, coder etc.

→ **Binary Attribute** - nominal attribute with only two categories or states i.e. '0' or '1'. where '0' means absent and '1' means present. If these values are 'true' and 'false', it's called 'Boolean'.
eg:- patient smokes - 0 or 1

i) **Symmetric** - if both states are equally valuable and carry same weight. eg:- Gender - male & female

ii) **Asymmetric** - if states do not carry same weight.
eg:- outcome of medical test.

→ **Ordinal Attribute** - attribute with +ve values that have meaningful order or ranking, but magnitude b/w successive values is not known. eg:- Grades for a student A, B, C, D, E

→ **Numeric attribute** - attribute with numeric values

→ **Interval scaled attribute** - numeric attributes measured in fixed and equal units. eg:- Calendar: 2010 - 2002 = 8
temp. $20^{\circ} - 15^{\circ} = 5^{\circ}$

→ **Ratio scaled attribute** - numeric value with inherent zero point

→ **discrete attribute** - has countably infinite set of values
colour, shape, pizza size, flavour

→ **Continuous** - not discrete

$$\text{Mean- } \bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

$$\text{Weighted Mean- } \bar{x}_w = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{w_1 + w_2 + w_3 + \dots + w_N}$$

Median = $L_1 = \left(\frac{N/2 - (\sum \text{freq})_x}{\text{freq}_{\text{median}}} \right) \text{width}$; L_1 = lower boundary
 N = No. of values in dataset
 $(\sum \text{freq})_x$ = sum of freq. of all intervals lower than median interval
 $\text{freq}_{\text{median}}$ = frequency of median value

Mode = data (value) with highest frequency

$$\text{mean-mode} \approx 3 \times (\text{mean-median}) \quad 1-4 = -3$$

$$\text{Standard deviation } s \text{ or } \sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

$$\text{Variance } v = s^2 \text{ or } \sigma^2$$

$$\text{Mean deviation } = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

$$\text{Skewness } = \frac{\bar{x} - \text{mode}}{s}$$

$$\text{Coefficient of Variation } = \frac{s}{\bar{x}} \times 100$$

$$\frac{1-4}{-3} = -3$$

Classification of DMS:-

- i) Based on mind databases
 - ↳ Relational
 - ↳ Transactional
 - ↳ Object Relational
 - ↳ Data Warehouse
- ii) Based on type of knowledge mind
 - ↳ Characterization
 - ↳ Discrimination
 - ↳ Association & Correlation Analysis
 - ↳ Classification
 - ↳ Prediction
 - ↳ Outlin Analysis
 - ↳ Evolution
- iii) Based on techniques used
 - ↳ ML
 - ↳ Statistics
 - ↳ NN
 - ↳ Pattern Recognition
 - ↳ DW oriented
- iv) Based on applications adapted
 - ↳ DNA
 - ↳ Finance
 - ↳ Telecommunications
 - ↳ Stock Markets
 - ↳ Email.

* DM task primitives :-

- ↳ A data mining task is represented in the form of a DM query
- ↳ DM query is defined in terms of DM task primitives
- ↳ DM task primitives allow user to interactively communicate with DM system.
- ↳ They are:-
 - i) set of task relevant data to be mined → attributes
 - ii) specify the kind of knowledge to be mined
 - iii) background knowledge to be used in KDD process
 - iv) interestingness measures and threshold for pattern evaluation
 - v) expected representation for visualizing the discovery.

* Integration of DMS with DBMS/DWS :-

- i) No coupling:- DMS DBMS
 no use of function
 → no commⁿ with DBMS, but with other storage media
- ii) Loose coupling:- will use some of the functionalities ^{some extent}
 suitable for small data sets, better than no-coupling
- iii) Semitight coupling - almost completely linked to DB
- iv) Tight coupling - DMS 100% linked to DBMS/DWS
 DBMS/DWS becomes part of DMS due to integration
 efficient & optimized implementation of DM.

* Factors of Data Quality

- i) **Timeliness**:- must be available within a timeframe to allow it to be useful for decision making.
- ii) **Believability**:- data values must be within range of possible results in order to be useful for decision making.
- iii) **Value added**:- data must provide additional value in terms of info. that offsets the cost of collecting and accessing it.
- iv) **Interpretability**- data must not be so complex that the effort to understand the info it provides exceeds the benefits of its analysis.
- v) **Accessibility**:- data must be easily accessible
- vi) **Accuracy**.
- vii) **Completeness**- there must be no missing values
- viii) **Consistency**-

* Data Preprocessing:- process of transforming raw data into an understandable format

Major tasks in Data Preprocessing

- i) **Data cleaning** - fill in missing values, smooth noisy data, identify and remove outliers, resolve inconsistencies
- ii) **Data Integration** - integration of multiple dbs, cubes, files
- iii) **Data Reduction** - to get compressed representation to dataset that produce same result as the original
- iv) **Data Transformation and data discretization**

* Data Integration :- combining data from multiple sources into a single coherent data source.

↳ Issues in data integration

- i) Schema integration :- metadata from different data sources must be integrated to match up equivalent real-world entities. This is called entity identification problem. Eg:- cust-id in one DB = cust-no. in another DB?
- ii) Handling Redundant data :- derived attributes may be redundant and inconsistent, leading to the same in result. duplications may occur and need to be detected & resolved.
- iii) Detection & Resolution of data value conflicts :- dif. in rep., scaling or encoding may cause the same real-world entity attribute values to differ in data sources being integrated.
metric v/s British units

↳ Handling Redundancy in Integration of Data (Base)

- i) Correlation Analysis :- for nominal data, a correlation relⁿ b/w two attributes A & B can be discovered by χ^2 (chi-square test) or Pearson-chi statistic

Suppose A has c distinct values $a_1, a_2, \dots, a_c$
B has r " " $b_1, b_2, \dots, b_r$

The data tuples described by A & B can be shown as Contingency table - values of A making columns & values of B making rows

A ↓ \ B →	B			
	b_1	b_2	...	b_r
a_1	11	12	...	$1r$
a_2	21	22	...	$2r$
$\vdots$	...	...	...	...
a_c	$c1$	$c2$	...	c_r

degree of freedom = $(r-1)(c-1)$

no. of cells = cr

Let $(A_i; B_j)$ = joint event that attribute A takes value A_i & attribute B takes value B_j

Let Then, $\chi^2 = \sum_{i=1}^C \sum_{j=1}^Y \frac{(O_{ij} - e_{ij})^2}{e_{ij}}$ for nominal data

where O_{ij} = observed frequency of (A_i, B_j)
 e_{ij} = expected frequency of (A_i, B_j)

$$e_{ij} = \frac{\text{count}(A = a_i) \times \text{count}(B = b_j)}{n}$$

n = no. of data tuples.

Larger the value of χ^2 , more likely the variables are correlated.
for numeric data, Covariance

$$\chi^2 \text{ (Pearson's Product Moment Coefficient)} = r_{AB}$$

$$= \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A\sigma_B} = \frac{\sum_{i=1}^n (a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A\sigma_B}$$

where n = no. of data tuples

$\bar{A}$ = Mean of A

$\bar{B}$ = Mean of B

σ_A = SD of A

σ_B = SD of B

$\sum (a_i b_i)$ = sum of AB cross product.

$$-1 \leq r_{AB} \leq +1$$

if $r_{AB} > 0 \Rightarrow A \& B$ are +vely correlated

if $r_{AB} = 0 \Rightarrow A \& B$ are independent

if $r_{AB} < 0 \Rightarrow A \& B$ are -vely correlated

Q.

Gender	male	female	Total row
Fiction	250 (90)	200 (360)	450
Non-fiction	50 (210)	1000 (840)	1050
Total (col ⁿ)	300	1200	1500

$$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840}$$

$$= 284.44 + 121.90 + 71.11 + 30.48 = 507.93$$

9.

A	B
20	8
12	34
9	4

$$\bar{A} = 13.667, \sigma_A = 5.686$$

$$\bar{B} = 15.333, \sigma_B = 16.289$$

$$n = 3$$

$$r_{A,B} = \frac{(20-13.667)(8-15.333) + (12-13.667)(34-15.333) + (9-13.667)(4-15.333)}{2 \times 5.686 \times 16.289}$$

$$= -0.1331706526 < 0$$

= -vely. correlated

* Data Reduction:- Volume of data is reduced in such a way that it produces the same output, ^{maintains integrity} and make analysis easier.

i) Dimensionality reduction - remove unimportant attributes

a) Wavelet Transform:- decomposes signal to dif. freq subbands used for image compression, allows natural clusters to become more visible. $O(N)$

b) Principal Component Analysis (PCA):- data is normalized and its orthogonal vectors are computed, then sorted in decreasing order of significance, lowest values are removed.

c) Attribute Subset Selection / Attribute Creation:- irrelevant attributes are removed to get subset of remaining attributes, out of which that is selected which has highest probability distribution.

ii) Numerosity Reduction:- replace original data volume by alternative smaller forms of data representation.

It is parametric if a model is used to estimate data
eg:- Regression and Log-linear models
Otherwise non-parametric (Histograms, clustering, Sampling, data cube aggregation)

- iii) Data Compression:- transformations are applied to get reduced & compressed form of org. data.
- i) Lossless - when reverse transformation is possible
 - ii) Lossy - " " " " " possible approx

* Data Transformations:- data is converted into appropriate forms for DM.

- i) Smoothing $\rightarrow$ Binning, Regression, Clustering
 - ii) Attribute Construction $\rightarrow$ new features are added
 - iii) Aggregation $\rightarrow$ Data is combined into a cube to remove redundancy
 - iv) Normalization $\rightarrow$ attribute data are scaled to fall in small range
- a) Min-Max

Q. $\rightarrow 200, 300, 400, 600, 1000$ normalised value

given $\begin{cases} \min = 0 \\ \max = 1 \end{cases} \Rightarrow V_i' = \frac{V_i - \min}{\max - \min} (1 - 0) + 0 = [0, 0.125, 0.25, 0.6, 1]$

$\therefore V' = \frac{V - \min}{\max - \min} (\text{new-max} - \text{new-min}) + \text{new-min}$

b) Z-score : $V' = \frac{V - \mu}{\sigma} = \frac{V - 500}{316.227766}$

$\Rightarrow [-0.949, -0.632, -0.316, 0.316, 1.581]$

c) Decimal Scaling :- $V' = \frac{V_i}{10^j}$; $j = \text{smallest integer such that } \max_i \left(\frac{V_i}{10^j} \right) \leq 1$

$\Rightarrow \left[\frac{200}{10^3} \leq 1, \frac{1000}{10^3} \leq 1 \right] \Rightarrow j = 3 \Rightarrow V' = \frac{V_i}{10^3}$

$\Rightarrow [0.2, 0.3, 0.4, 0.6, 1]$

* Data discretization :- transform numeric data by mapping values to interval or concept labels.

i) Binning (top-down split, unsupervised)

Q. 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

a) Equal width (distance)

partitioning
 $(34-4)/3 = 10$

Bin 1: 4, 8, 9, 15

Bin 2: 21, 21, 24, 25

Bin 3: 26, 28, 29, 34

(b) Equal freq. (depth)

partitioning
 depth = 4

Bin 1: 4, 8, 9, 15

Bin 2: 21, 21, 24, 25

Bin 3: 26, 28, 29, 34

5, 10, 11, 13, 15, 35, 50, 55, 72, 92, 204, 215

a) $(215-5)/3 = 70$

Bin 1: 5, 10, 11, 13, 15, 35, 50, 55, 72

Bin 2: 92

Bin 3: 204, 215

(b)

Bin 1: 5, 10, 11, 13

Bin 2: 15, 35, 50, 55

Bin 3: 72, 92, 204, 215

ii)

Clustering (top-down split or bottom-up merge, unsupervised)

K-means

Bin 1: 5, 10, 11, 13, 15, 35

Bin 2: 50, 55, 72, 92

Bin 3: 204, 215

iii) Decision Tree (top down split supervised)

has class labels, use entropy to find split point

iv) Correlation Analysis :- ChiMerge: χ^2 -based
 bottom up merge
 uses class info.

v) Concept Based Hierarchy Generation for Nominal Data :-
 Organize concepts hierarchically
 associated with each data in DW
 facilitate drilling & role in DW to view data in multiple
 granularity.

- by specifying partial ordering of attributes
street < city < state < region < country
- by specifying a portion of hierarchy by explicit data grouping
(Urbana, Champaign, Chicago) < Illinois
- by specifying only a partial set of attributes
street < city (not others)
- based on no. of distinct values per attribute
country(5) > state(365) > city(3567) > street(6M)

Grade	Rep	Dem	Ind	Σx
M	200(180)	150(180)	50(40)	400
F	450(270)	300(270)	50(60)	600
C	450	450	100	1000
DF = (r-1)(c-1) = (2-1)(3-1) = 2				
$E_{rc} = (n_r \cdot n_c) / n$				
$\chi^2_{cal} = \frac{(200-180)^2}{180} + \dots$				
= 16.2				
$\chi^2_{tab} = 10.597$				
if $\chi^2_{cal} > \chi^2_{tab} \rightarrow$ reject				
if $\chi^2_{cal} < \dots \rightarrow$ accept				

MODULE-2



Page No:

Date: .../.../...

DW: Datawarehouse is a subject oriented, integrated, non-volatile and time variant collection of data in support of management's decision.

i) Subject Oriented- data is stored by subjects not application

Order Billing
Customer Billing
Student Marks

Sales

Account

Institute Rank } Evaluation

ii) Integrated - data in DW comes from dif. operational systems. This is good to remove inconsistencies.

iii) Time variant - DW stores current as well as historical data. Every data structure in DW allows

(a) analysis of Past (b) relate to present (c) forecast future

iv) Non-Volatile - data is never updated / deleted from DW

v) Data Granularity - In DW, it is efficient to keep data summarized at different levels

OLTP (Online Transaction processing)
operational

DW
OLAP (Online Analytical processing)
Informational

Characteristics

processing

Oriented to

User

Uses

Access

Metric

Priority

No. of records accessed

Size

Focus

Query

View

transaction, application
Clerk

Day to Day

Read + Write

transaction throughput

high performance & availability

10 - 100

GB - to high order of GB

data in

Short & Simple

detailed, flat, summarizes relational

analysis, subject

Data Analyst

Decision Making, Prediction, Planning

Mostly Read

query & response time

high flexibility, end user autonomy

millions

> TB

info. out

Complex

summarized, multidimensional

→ improve efficiency & accept complexity

Data Modelling - process of designing schema of detailed & summarized info. of DW. It has 3 levels

- i) Conceptual - explains semantics of data.
- ii) Logical - defines data in structural form
- iii) Physical - describe how data is rep. in DB.

Data Warehouse Models

- i) **Enterprise Model** :- contains all info. about subjects ^{time taking to build} related to entire organization (detailed & summarized)
- ii) **Data Mart** - subset of enterprise model / corporate wide data, has value to a specific group of users. limited scope. It is summarized. Can be dependent if sourced directly from enterprise DW or independent if captured from one or more operation systems or external providers like from government.
- iii) **Virtual Warehouse** :- contains data copied from multiple sources during production.

	Top down DW development	Down Top DW development
Pros	systematic solution for enterprise minimizes integration problems	flexibility, low-cost, rapid-return of investment
Cons	expensive takes long time to develop lacks flexibility difficulty in achieving consistency	integration problems

Data Cube - integrated model of data, that allows viewing of data of interest. it is defined by facts & dimensions.

LP

Page No:

Date: .../.../...

Dimensional Modelling:- data in warehouse are usually multi-dimensional having measure and dimension attributes. values can be measured & aggregated like sum, min, max, avg. which major attributes and their summary funⁿ work upon.

i) Facts :- numerical measures by which one can analyse relationships b/w dimensions. Relations having such multidimensional data are fact table.

Books

BID	tid	Num
B1	1	25
B2	2	16

Sum = 41

ii) Dimensions :- collection of logically related attributes and is viewed as an axis for modelling the data. Dimension table is a table associated with each dimension helps in describing the dimension further.

Bid	Author	Price
B1	ABC	40
B2	XYZ	70

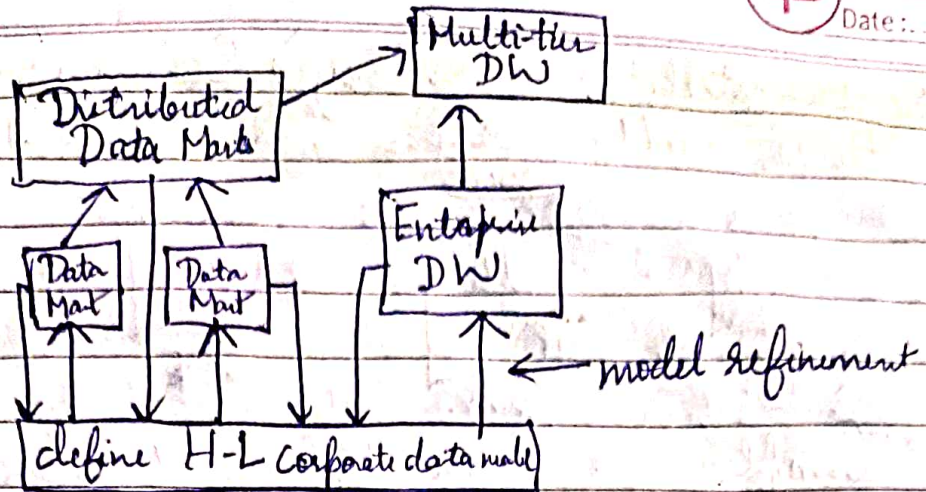
explains Bid

tid	address	Date

explains tid

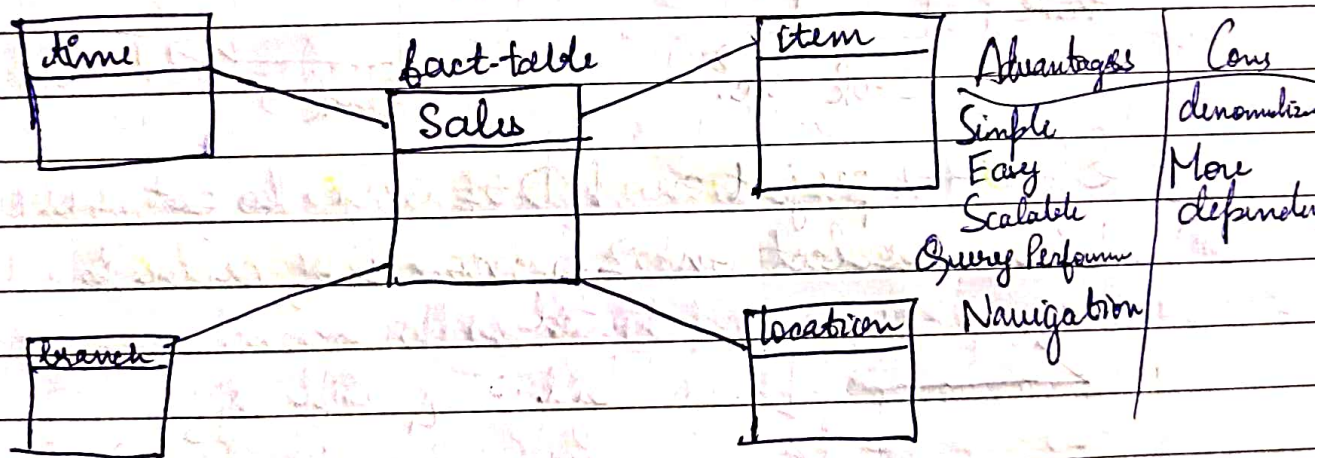
Recommended Approach for DW development

- high level corporate model is defined within a short period (1-2 months) that provides entire data of corporation/enterprise
- independent data marts can be implemented in parallel with enterprise warehouse.
- distributed data marts can be constructed to integrate dif. data marts via hub servers.
- Multi-tier data warehouse is formed.



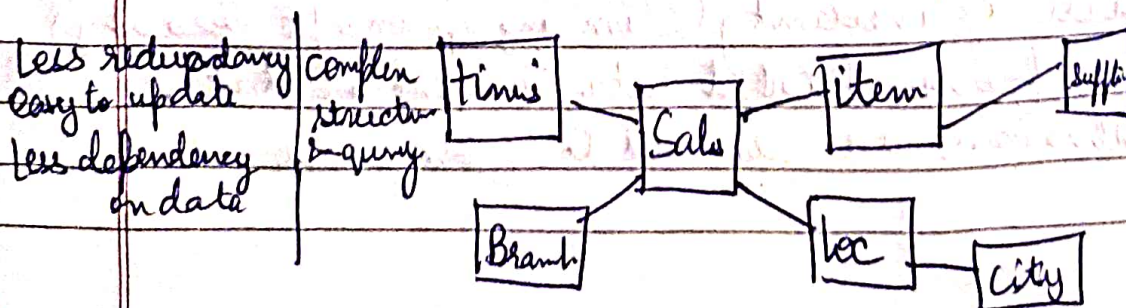
Schemas / Conceptual Modeling of Multidimensional Data Models of Data Warehouses

- i) **Star Schema**:- DW has large central fact table containing bulk of data with no redundancy and a set of smaller attendant tables (dimension tables), one for each tab/dimension

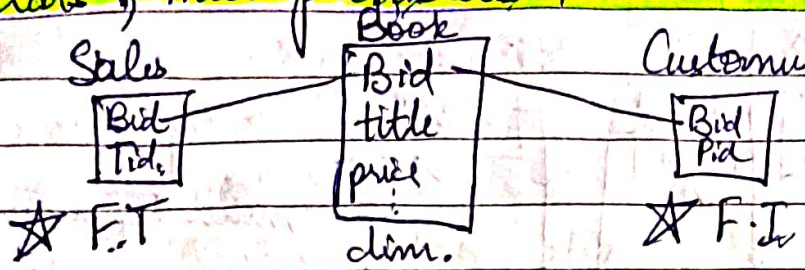


1) **not normalized** Simplest & Easiest, optimizes navigation through DB, most suitable for query processing, More dependencies

- ii) **Snowflake Schema**:- variant of star, some dim. table are normalised, thus there is further splitting of data into more tables



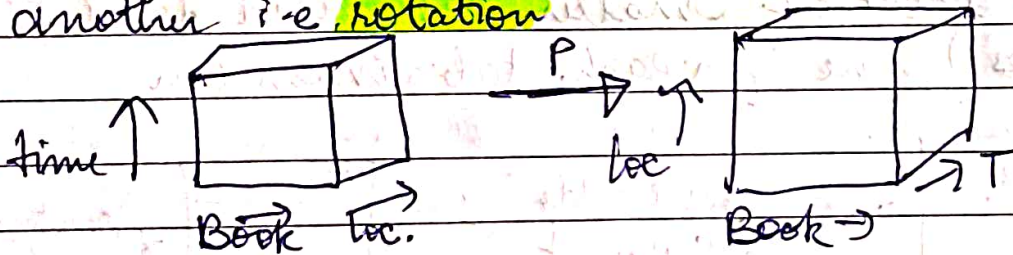
iii) **Fact Constellation (Galaxy) Schema**:- Collection of stars; multiple factors share dim.



Complex due to multiple fact table
Dim. Tables are very large
difficult to manage.

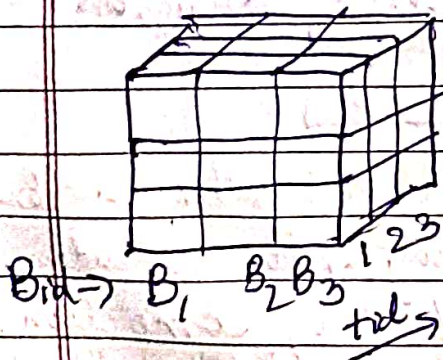
✓ OLAP Operations

i) **Pivoting**:- technique to changing from 1-D orientation to another i.e. rotation



ii) **Slice**:- project on 1-D of cube to get subcube

iii) **Dice**:- select on 2 or more D of cube to get subcube



bid	L ₁	L ₂	L ₃	
B ₁	25	12	22	→ 59
B ₂	11	23	12	→ 46
B ₃	16	17	12	→ 45

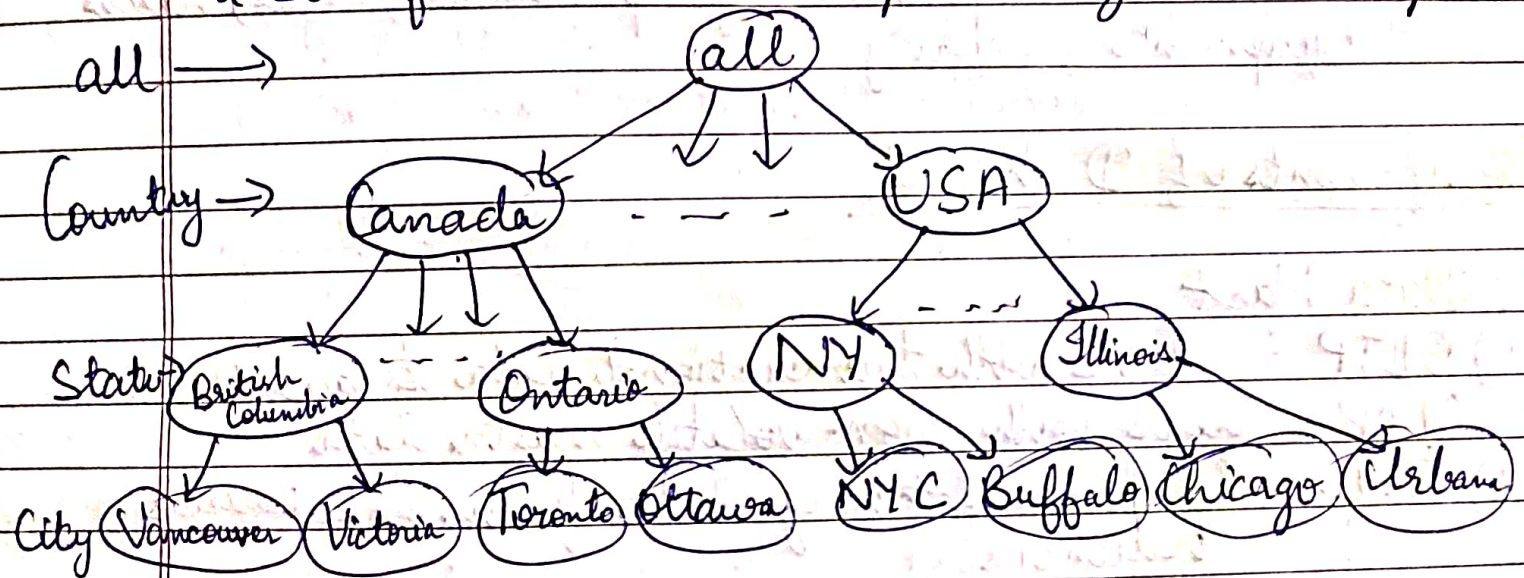
iv) **Roll up / drill up**:- convert data with fine granularity to coarse granularity (climbing up concept hierarchy)
navigate from detailed data to general data.
Dimensions are removed from cube

- v) **Drill down** :- convert data with coarse granularity to fine granularity (stepping down concept hierarchy) navigates from less detailed data to more detailed data. Dimensions are added into the cube.
- vi) **Drill Across** - execute queries involving more than one table
- vii) **Drill through** - use relⁿ db SQL facilities to step down to bottom of data cube down to back-end relational tables.

Incremental

Updating

Concept Hierarchy :- defines a sequence of mapping from a set of low level concepts to high level concepts.



Advantages → Data Cube ← **Disadvantages**

summarized view of data
improved performance of data
Quick & Better analysis
Accessibility
Decision Making
Navigation

required specialized
training expertise
Complex to set up

Data Warehouse Implementation :- series of activities that are needed to create a fully functional DW.

It has various phases:

- i) Planning :- helps in getting pathway to follow to achieve our goals
- ii) Data Gathering :- collection of data from various sources for analysis and reporting, Time consuming as we need to identify what is going to be helpful for the organisation
- iii) Data Analysis - generating meaningful insights for decision making
- iv) Business Actions - activities based on decisions made to make profit. Eg:- production, sales, loans, equipments, security

Components of D W I

- i) Data Mart
- ii) OLTP - deals with transactional data frequently changing
- iii) OLAP - deals with non-volatile master data
- iv) ETL - helps in fetching data from dif. sources in warehousing
↳ Extraction, Transformation & Loading
↳ extract → clean → transform → load → refresh
↑
- v) Metadata Repository - helps in getting granularity of data

Goals of DWI

- ↳ Creating & Managing efficient data repository
- ↳ Secured & easy access of info. to users
- ↳ Consistent info to be provided
- ↳ Quick & effective decision making
- ↳ Clean data that is required by analyst
- ↳ Adaptive, Accurate, Verified.

	2 Tier Architecture	3-Tier Architecture	4-Tier Architecture
Client	Client - Server	Most Widely Used	i) Database
Server	Not expendable	i) Bottom layer - after cleaning, transformed data is loaded	ii) Application
DB	Easy to Maintain	ii) Middle Layer: has business logic, OLAP server	iii) Presentation
Req.	Fast Communication	iii) Top Layer: Query tools, Front-end Client Layer	iv) Client
Res.	Doesn't support large no. of end users		
ROLAP			
MOLAP			
HOLAP			

